How Many Security Stations Does the Airport Need?

CS166 First Project: Airport Security Queues

September 2026

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Videos: Almira (Google Drive) — Tara (Loom)

Notebook: All numbers, tables, and figures in this report come from the submitted Python notebook, which reproduces them from start to finish.

1 Introduction

Every traveler at a large airport passes through security, and the airport has to decide how many screening stations to keep open. With too few, lines grow without limit; with too many, officers stand idle. In this project we model the checkpoint as a set of parallel queues, simulate it in Python, and combine the simulation with queueing theory to answer one question: *how many service stations should airport security have?*

We first built a basic model with k stations and checked it against the two test cases we were given. We then added a single senior security officer who inspects the 3% of travelers flagged for additional screening. That officer turns out to dominate everything else. The officer raises the number of stations needed to keep lines from growing without limit from 6 to about 8 and causes almost all of the waiting that remains.

The report moves from the model and its verification to theory, simulation results, the gap between the two, and a staffing recommendation.[1]

2 Model

2.1 Assumptions

1. **Arrivals.** Travelers arrive independently at a constant mean rate λ, 24 hours a day. The number of arrivals in any interval is therefore Poisson distributed and the gaps between arrivals are exponential with mean $1/\lambda$. This is why the simulation schedules each next arrival by drawing an Exp(λ) gap.

2. **Choosing a line.** On arrival, a traveler joins the line with the fewest people *in it*, counting the traveler currently at the station, and never switches lines. Ties go to the lowest-numbered station.

[1] #professionalism: the report follows a Model–Implementation–Theory–Results–Conclusion structure rather than answering the assignment prompts one by one. All quantities carry units (minutes, travelers per minute, people), and every reported number is reproducible from the notebook with a fixed seed.

3. **Stations.** Each line has one station that screens one traveler at a time, first come, first served (FIFO). Everyone behind that traveler waits.

4. **Regular screening.** Screening times are independent draws from a normal distribution $\mathcal{N}(\mu_1, \sigma_1^2)$ truncated at zero, so that no time is negative.

5. **Additional screening.** After regular screening, each traveler independently needs additional screening with probability p_s. That traveler stays at the station, which is then *blocked*, until the senior officer clears them.

6. **The senior officer.** There is one officer, who handles one traveler at a time, in the order in which stations request help. Inspection times are normal $\mathcal{N}(\mu_2, \sigma_2^2)$ truncated at zero.

Truncation raises the mean. We treat μ and σ as the parameters of the untruncated normal, as the second test case does. The true mean regular screening time is then 30.04 s, not 30 s, and the true mean officer inspection is 2.575 min, not 2 min (Appendix A). The small shift matters: with $k = 5$ stations, utilization is $\lambda E[B]/k = 1.0015$, so five stations are unstable even though "30 seconds per traveler" suggests exactly full capacity.

Table 1: Model parameters. All times are in minutes; each parameter is set once, in the notebook's parameter cell.

Symbol	Meaning	Value	True mean after truncation
λ	arrival rate	10 travelers/min	–
μ_1, σ_1	regular screening, normal before truncation	0.5 min, 1/6 min	0.5007 min
p_s	probability of additional screening	0.03	–
μ_2, σ_2	officer inspection, normal before truncation	2 min, 2 min	2.575 min
k	number of stations	varied, 6–16	–

2.2 State variables and update rules

The state consists of the simulation clock; for each station, the ordered list of waiting travelers (with their join times) and the station status (*idle*, *serving*, or *blocked*); and, for the officer, a busy flag and the ordered list of blocked stations waiting for inspection. The state changes only at five kinds of events:

1. **Arrival at time t.** The traveler joins the line with the fewest people in it. If that station is idle, screening starts at t. The next arrival is scheduled at $t + \text{Exp}(\lambda)$.

2. **Start of screening.** The first waiting traveler moves to the station, which becomes *serving*. Their waiting time is recorded, and the end of screening is scheduled after a draw from the regular screening distribution.

3. **End of regular screening.** With probability $1 - p_s$ the traveler leaves, the station becomes idle, and the next traveler in that line, if any, starts at once. With probability p_s the station becomes *blocked* and requests the officer.

4. **Start of an inspection.** If the officer is free, they take the first blocked station in their list and schedule the end of the inspection after a draw from the officer's distribution.

5. **End of an inspection.** The traveler leaves, the station becomes idle and starts its next traveler, and the officer moves on to the next blocked station.

2.3 Metrics

The **waiting time** of a traveler is the time from joining a line to the start of regular screening. A traveler is counted if their screening starts within the measured period. The **queue length** of a line is the number of people waiting, *excluding* the traveler at the station; we report its time average per line. The **maximum queue length** is the largest number waiting in any single line during a simulated day.[2]

3 Implementation and Verification

3.1 Code structure

The simulation is event driven: the clock jumps from one scheduled event to the next (Figure 1). `Schedule` holds the clock and a heap of pending `Event` objects, so the next event is always found in $O(\log n)$ time. Each event also carries a sequence number, so events scheduled for the same instant run in the order they were scheduled. Model objects interact only through this calendar: each method schedules future events, and `Schedule.run_next_event` advances the clock and calls the stored method. `Airport` owns k `Queue` objects and one `SeniorOfficer`. Arrivals, regular screening times, referral decisions, and inspection times each come from a separate seeded random stream. The function `run_simulation` builds and runs one replication, and `run_trials` repeats it over independent seeds.[3]

Changes for the extended model. We added the officer with four local changes (the orange branch in Figure 2). First, a new `SeniorOfficer` class holds the officer's state and a first-come, first-served list of blocked stations. Second, `Queue.finish_service` now branches: with probability p_s it marks the station as blocked and requests the officer instead of releasing the traveler. Third, a new method, `Queue.finish_additional_screening`, releases the station when the officer is done. Because a blocked station stays busy, nobody else in that line can start screening, while other lines carry on. Fourth, `Airport` owns the officer and the referral stream. Since the referral decisions and inspection times use their own random streams, switching the officer off leaves every other random draw unchanged.

3.2 Verification

We checked the code at three levels (notebook §5–7).

Watching it run. A short event trace and Figure 3 show arrivals joining the emptiest line, idle stations preferred, and a blocked station holding its line while the others keep working.

Hand-computed cases. With deterministic inputs, the code reproduces paper calculations exactly, including blocking and two stations sharing one officer.

Checks against theory and the test cases. Travelers are conserved, and Little's law holds to within 0.2%. With one station the simulation matches the Pollaczek–Khinchine formula for deterministic, exponential, and truncated normal service at utilizations from 0.3 to 0.9. When deliberately overloaded, the backlog grows at 2.07 travelers/min against a predicted 2.00. Both provided test

[2] #modeling: we separate assumptions, parameters, state variables, and update rules, so the model can be reimplemented from this section alone. We tie the event generation to the Poisson–exponential relationship and state exactly what each metric counts. Section 8 discusses how realistic each assumption is.

[3] #cs110-CodeReadability: every class and method has a docstring stating its purpose, inputs, and units. All parameters live in one cell, with no magic numbers elsewhere. Debug printing sits behind a `verbose` flag, and invalid inputs (for example $\rho \geq 1$ in a formula) raise explicit errors.

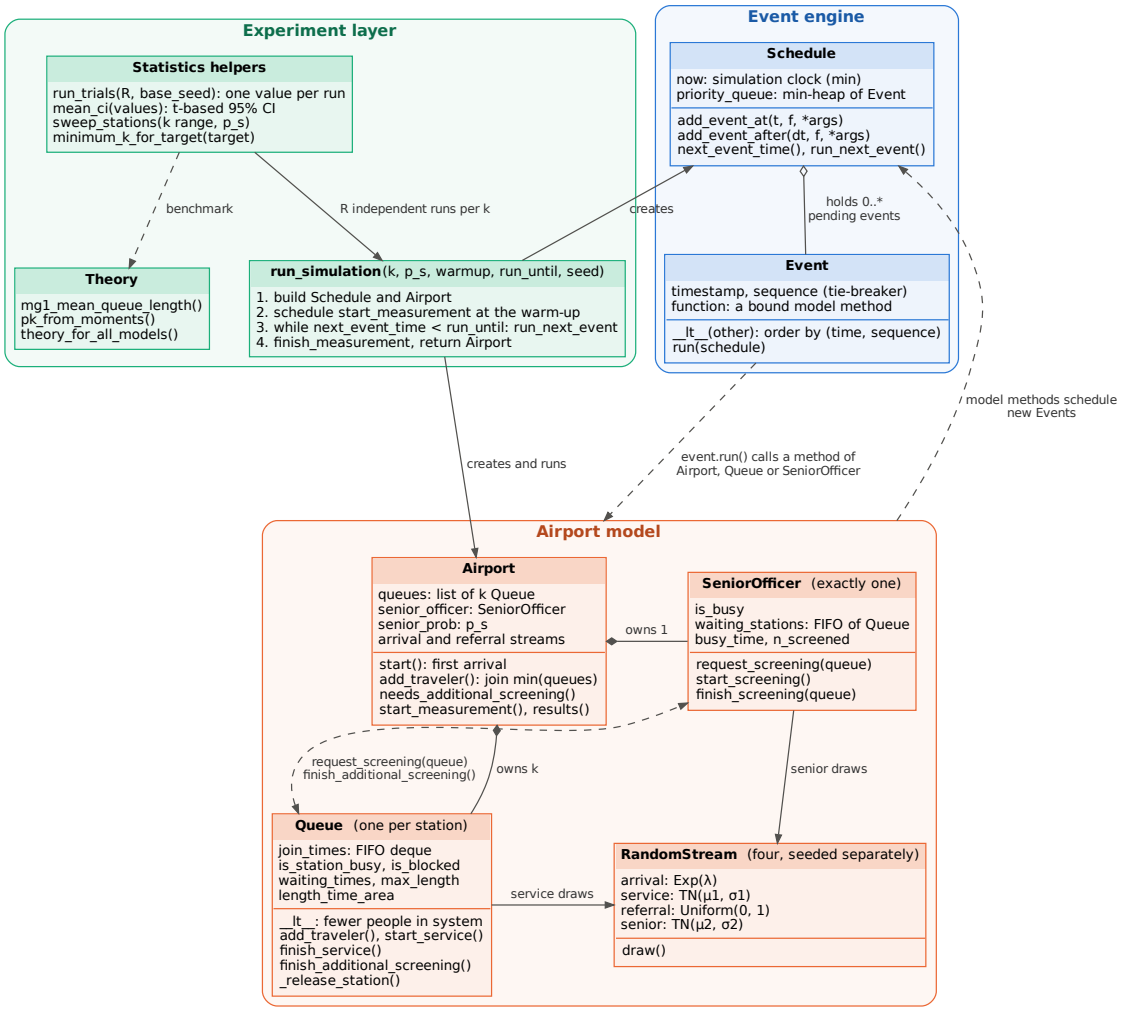


Figure 1: Structure of the simulation code. Boxes list the main state variables above the line and the main methods below it. Solid arrows denote creation, ownership, or random draws; the filled diamond marks composition, and the open diamond marks the pending events held by `Schedule`. Dashed arrows denote calls made through the event calendar and the exchange between a blocked station and the senior officer.

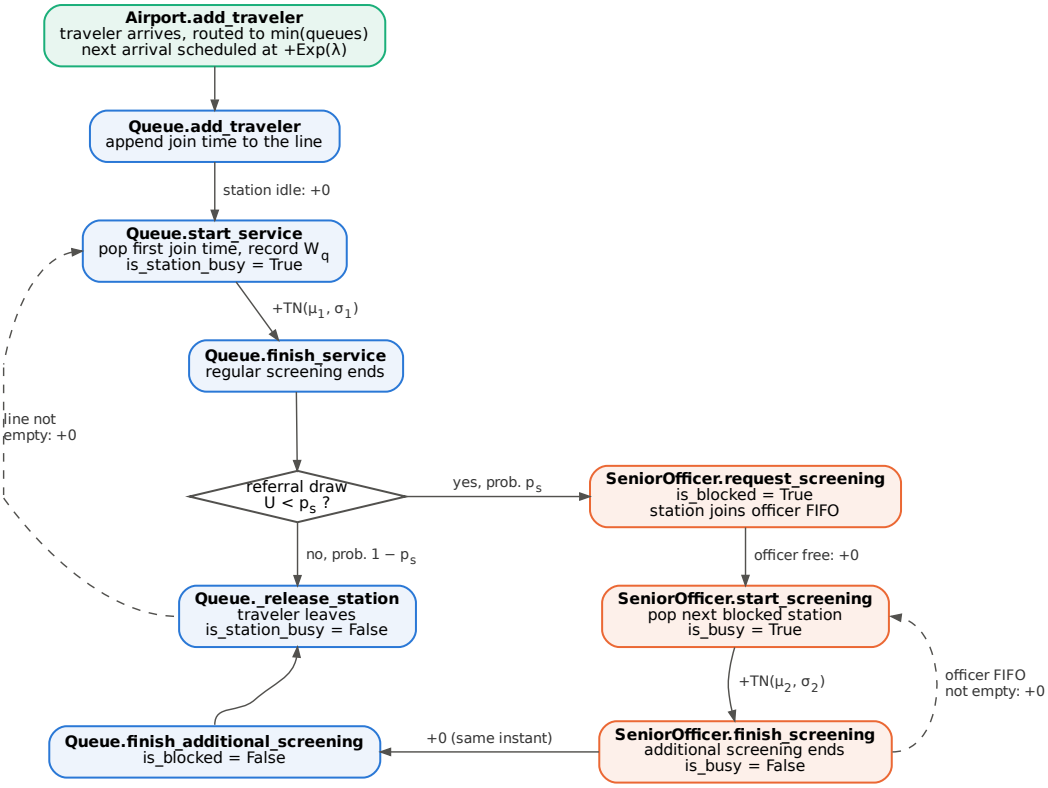


Figure 2: Sequence of events for one traveler. Blue boxes are `Queue` methods and orange boxes are `SeniorOfficer` methods. Each edge label gives the delay after which the next event is scheduled: +0 for the same instant, otherwise a random draw from the indicated distribution. Dashed edges hand control to the next waiting traveler.

cases fall inside their target bands (Table 2): 0.457 ± 0.012 against 0.463 ± 0.025, and 0.555 ± 0.021 against 0.566 ± 0.025 people per line. Setting $p_s = 0$, or $\mu_2 = \sigma_2 = 0$ with $p_s = 0.03$ or even $p_s = 1$, reproduces all 2,085 individual waiting times of the basic model exactly.

One check changed our model. If "shortest line" counts only the people waiting, idle and busy stations both look empty, travelers queue behind busy ones, and test case 1 gives 0.802 ± 0.023, far off target. Counting the traveler at the station, as in Section 2, fixes this.[4]

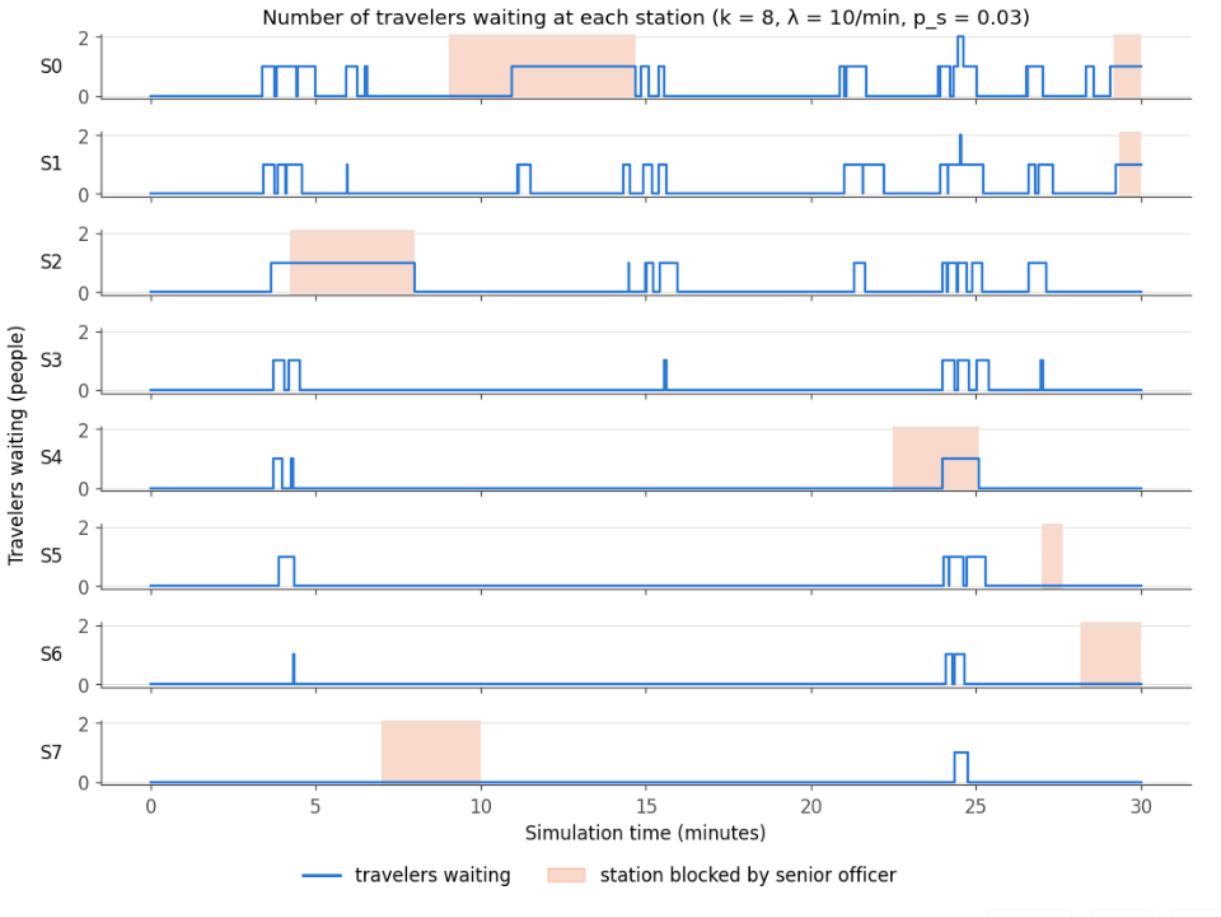


Figure 3: State of one 30-minute run of the extended model with $k = 8$ stations. Each panel shows the number of travelers waiting at one station, not counting the one being screened; orange spans mark periods when the station is blocked by an officer inspection. Shortest-line routing keeps the lines within about one person of each other, and a line grows mainly while its station is blocked.

Stationarity. We checked this empirically by averaging 200 runs from an empty airport and comparing consecutive time windows (notebook §6.1). Set 1 levels off within about 15 minutes and set 2, which is closer to saturation, within about 60, so the suggested warm-ups of 100 and 200 minutes are sufficient. The model with the officer settles much more slowly, so we checked it separately at $k = 8$ and $k = 9$ and use a 300-minute warm-up there (Appendix B).

[4] #cs166-PythonImplementation: correctness is demonstrated with a state trace and visualization (Figure 3), deterministic hand-checked cases, comparisons with theory, both provided test cases, and exact reduction checks for the extended model. The notebook runs top to bottom and reproduces every result.

Table 2: Mean number waiting per line in the two provided test cases: 200 independent runs each, with 95% confidence intervals of the mean. The time average is taken over 400 min after the warm-up; the snapshot is a single instant at the end of each run, so its interval is wider.

	k	ρ	Target	Time average	End snapshot
Set 1	6	0.833	0.463 ± 0.025	0.457 ± 0.012	0.485 ± 0.091
Set 2	14	0.920	0.566 ± 0.025	0.555 ± 0.021	0.589 ± 0.081

4 Theoretical Analysis

We analyze the system in four steps, each adding one feature. Full derivations are in Appendix A.

Independent lines. If every traveler picked a line at random, each line would receive an independent Poisson stream of rate λ/k and behave as an M/G/1 queue. If S is the time one traveler occupies a station, the Pollaczek–Khinchine formula gives the mean wait

$$W_q(k) = \frac{\lambda E[S^2]}{2\,(k - \lambda E[S])}, \qquad k > \lambda E[S], \tag{1}$$

and, by Little's law, a mean queue length per line of $L_q = \lambda W_q/k$. The four steps differ only in the distribution of S.

Step 1: regular screening only. Here $S = B$ with $E[B] = 0.501$ min and $\mathrm{CV}_B^2 = 0.109$, so screening is nearly deterministic. Stability requires $k > \lambda E[B] = 5.01$, so at least 6 stations, and at $k = 6$ the predicted wait is 1.40 min.

Step 2: additional screening, with the officer always free. Now $S_0 = B + IA$, where $I \sim \mathrm{Bernoulli}(p_s)$ and A is the inspection time. The mean rises only to 0.578 min, but CV^2 jumps to 0.89. Six stations still suffice, yet the predicted wait at $k = 6$ grows tenfold, to 14.3 min.

Step 3: the officer as a queue. Referrals reach the officer at rate $\lambda_s = p_s\lambda = 0.3$ per minute, so the officer is busy $\rho_s = \lambda_s E[A] = 0.773$ of the time, *whatever the value of* k. Only 3% of travelers need the officer, but each inspection lasts five times as long as a regular screening. Treating the officer as an M/G/1 queue gives a mean wait for the officer of $E[W_s] = 6.03$ min (an exponential inspection time would give 8.75 min), with a standard deviation of 7.32 min from Takács' formula.

Step 4: blocking. A referred traveler occupies their station while waiting for the officer and while being inspected, so

$$S_{\text{eff}} = B + I\,(A + W_s), \qquad E[S_{\text{eff}}] = 0.759 \text{ min}. \tag{2}$$

Replacing W_s by its mean (case (a)) gives $\mathrm{CV}^2_{\text{eff}} = 3.92$; using its full variance (case (b)) gives 6.71. Little's law splits the capacity requirement into stations screening, stations holding a traveler under inspection, and stations blocked while waiting for the officer:

$$\lambda E[S_{\text{eff}}] = \underbrace{5.007}_{\text{screening}} + \underbrace{0.773}_{\text{inspection}} + \underbrace{1.810}_{\text{waiting for officer}} = 7.59. \tag{3}$$

The threshold is therefore about 7.6 stations, so at least 8 (9 if inspections were exponential). Waiting for the officer costs more station capacity than the inspections themselves. There is also a second constraint that no number of stations can relax: the officer alone must satisfy $\lambda p_s E[A] < 1$, and as $\rho_s \to 1$ the officer's wait, and with it the station requirement, grows without bound.[5]

Table 3: Approximate capacity requirement at each analytical step. The last three rows are approximate conditions for the coupled system, not proofs of stability.

Step	$E[S]$ (min)	CV_S^2	$\lambda E[S]$	Minimum k
Regular screening only	0.501	0.109	5.007	6
Officer always free	0.578	0.886	5.780	6
Officer as M/G/1, case (a)	0.759	3.923	7.590	8
Officer as M/G/1, case (b)	0.759	6.710	7.590	8
Officer as M/M/1 (sensitivity)	0.840	5.428	8.404	9

Maximum queue length. The daily maximum is an extreme value of a dependent process and has no closed form. Theory still predicts its behavior. Heavy-traffic tails scale with $\rho(1 + \mathrm{CV}^2)/(1-\rho)$, so with $\mathrm{CV}^2_{\mathrm{eff}}$ between 4 and 7 the worst line should be many times the mean. Below the threshold, backlogs grow over the day, so the maximum depends on the length of the day. And because one officer serves all stations, several lines can freeze at once, a shared shock that single-queue formulas ignore.

Approximations. The analysis assumes that each line receives an independent Poisson stream, as under random routing (A1); that referrals reach the officer as a Poisson stream (A2); and that screening time, inspection time, and officer delay are independent (A3). It also describes a long-run steady state, while each simulated day is finite. A1 and A2 should both make the predicted waits too high: shortest-line routing balances the lines, and a blocked station cannot send a new referral. Section 6 tests both.

5 Simulation Results

5.1 Experimental design

We simulate $k = 8$ to 16 stations, starting just above the threshold of 7.6 given by Equation (3). At $k = 6$ the model with the officer is clearly unstable: its backlog reaches about 730 travelers after 1,000 minutes. At $k = 7$ the evidence is inconclusive (Appendix B). Each replication discards a 300-minute warm-up and then measures one day of 1,440 minutes. We run 50 independent days at every k, for the basic model ($p_s = 0$) and the extended model ($p_s = 0.03$). Days with the same index share seeds across k and across models, which reduces noise in the comparisons. Each day contributes one value of each metric, and we report the mean with a 95% confidence interval of the mean, $\bar{x} \pm t_{0.975,49}\, s/\sqrt{50}$.[6]

[5] #cs166-TheoreticalAnalysis: we build the analysis up in steps, from M/G/1 lines to an officer queue coupled to the lines through blocking. We derive the capacity threshold and the waiting-time predictions and compare them with the simulation in Section 6, where a targeted experiment identifies why they disagree.

[6] #confidenceintervals: each interval uses one value per independent day, taken after the warm-up, and the t distribution. It describes uncertainty about the *mean* over days; it is not an interval for a single day, and we do not read it as "95% of days meet the target" (Section 7).

5.2 Waiting time, queue length, and the worst line

Table 4 and Figures 4–6 summarize the results.[7] In the extended model the mean wait falls from 2.99 ± 0.52 min at $k = 8$ to 0.09 ± 0.04 min at $k = 16$, with sharply diminishing returns: the ninth station saves 1.21 min per traveler, the sixteenth only 0.05 min. The basic model never waits more than 0.035 min on the same days, so the officer causes almost all of the waiting. The mean queue length per line follows the same curve, as Little's law requires: $L_q = \lambda W_q / k$ holds at every k to the precision shown.

The worst line of the day falls more slowly than the mean, from 24.8 ± 2.9 people at $k = 8$ to 2.2 ± 0.6 at $k = 16$. Relative to the mean queue it actually grows, from 6.6 to 38 times, so even when the average line is nearly empty, occasional long lines persist. They form mainly behind blocked stations, as in Figure 3. In the basic model the worst line is exactly one person on every simulated day from $k = 10$ onward.[8]

Table 4: Simulated metrics: means over 50 independent days per station count, with 95% confidence intervals of the mean. Queue length is per line; the worst line is the daily maximum of a single line.

	Basic model	Extended model		
k	Wait (min)	Wait (min)	Queue length (people)	Worst line (people)
8	0.035 ± 0.001	2.988 ± 0.518	3.736 ± 0.645	24.82 ± 2.88
9	0.015 ± 0.000	1.782 ± 0.351	1.981 ± 0.389	18.02 ± 2.27
10	0.006 ± 0.000	1.174 ± 0.265	1.176 ± 0.265	13.58 ± 2.02
11	0.003 ± 0.000	0.781 ± 0.202	0.712 ± 0.183	10.24 ± 1.71
12	0.001 ± 0.000	0.522 ± 0.155	0.436 ± 0.129	7.62 ± 1.45
13	< 0.001	0.346 ± 0.115	0.268 ± 0.088	5.58 ± 1.18
14	< 0.001	0.228 ± 0.083	0.164 ± 0.060	4.10 ± 0.96
15	< 0.001	0.147 ± 0.059	0.099 ± 0.039	2.96 ± 0.77
16	< 0.001	0.093 ± 0.041	0.058 ± 0.026	2.24 ± 0.57

6 Why Theory and Simulation Disagree

The theory gets the structure right. The officer, not regular screening, drives congestion; the officer's measured utilization is 0.778 ± 0.007 against a predicted 0.773, with 0.0301 ± 0.0003 of travelers referred against $p_s = 0.03$; and waits fall steeply just above 7.6 stations. The numbers, however, are far off. Case (b) predicts 54 min at $k = 8$, against a simulated 3.0 min. Even without the officer, the independent-line formula predicts 0.465 min at $k = 8$, while the simulation gives 0.035.

The two differ in how travelers choose a line and in how the officer's queue is modeled. To separate these, we reran the simulation with a single change: each traveler joins a *random* line (notebook §8.6). This is exactly the routing that the independent-line theory assumes. We also added a lower benchmark, one pooled line feeding all k stations (an M/G/c queue), using the Erlang C formula with the Allen–Cunneen correction for non-exponential service [1].

[7] #dataviz: every results figure shows means with 95% confidence intervals of the mean, overlays theory where it exists, and has a caption stating the data, the sample size, and what the error bars mean. Where one curve would flatten the others, a zoomed panel (Figures 4 and 5) or a log scale (Figure 7) keeps every series readable. Figure 3 visualizes the state of the simulation itself.

[8] #cs166-EmpiricalAnalysis: all three metrics are estimated from 50 independent days per condition, compared across k and between the two models with common random numbers, and interpreted in terms of travelers and

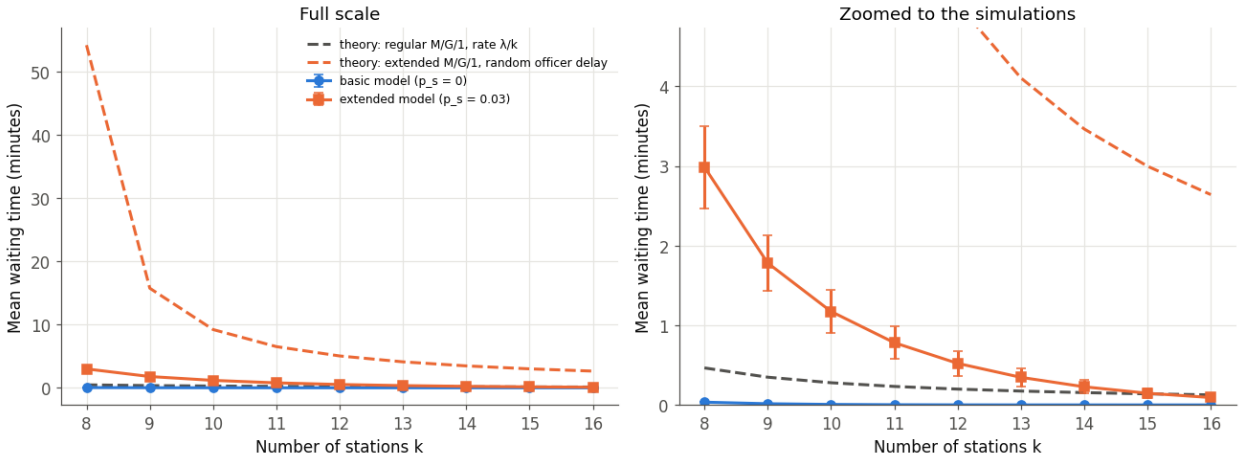


Figure 4: Mean traveler waiting time against the number of stations k, for the basic and extended models: means over 50 independent days with 95% confidence intervals of the mean. The dashed curves are the independent-line theory for regular screening only and for case (b) with the officer. The right panel zooms the vertical axis to the simulations; case (b) leaves the top of that panel at small k.

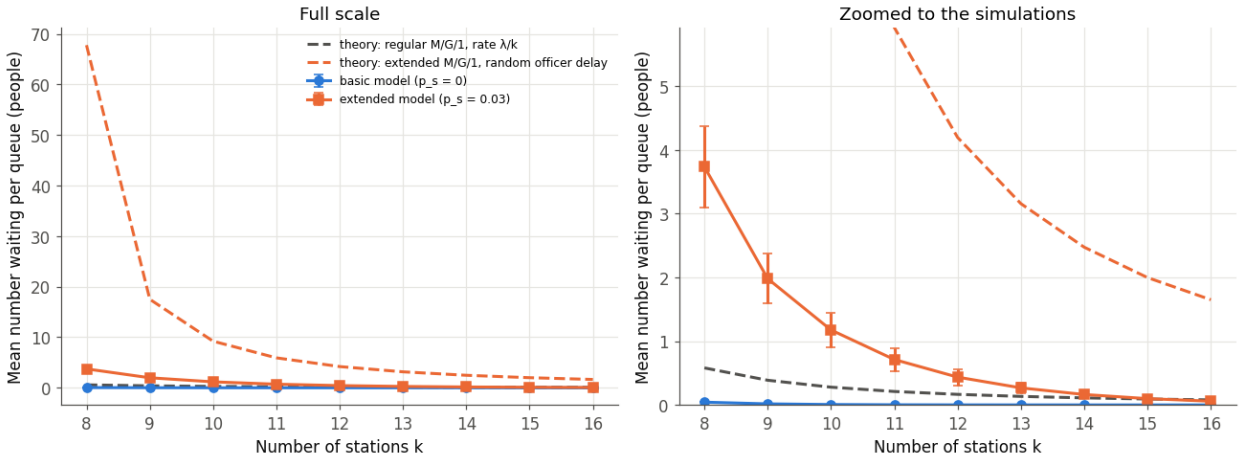


Figure 5: Mean number of travelers waiting per line against k, from the same 50 days as Figure 4. The theory curves are per line, $(\lambda/k)\,W_q$. The right panel zooms the vertical axis to the simulations. The shape mirrors Figure 4, as Little's law requires.

Table 5: Mean waiting time (minutes) under theory and under two routing rules in the simulation. Random routing: 30 days per k; shortest-line routing: the 50 days of Table 4. The pooled line is the M/G/c approximation with the case (b) service moments.

	Independent-line theory		Simulation		Pooled line
k	Case (b)	Case (a)	Random line	Shortest line	M/G/c
8	54.23	34.62	7.67	2.99	6.01
10	9.22	5.89	3.82	1.17	0.39
12	5.04	3.22	2.65	0.52	0.07
14	3.47	2.21	2.08	0.23	0.01
16	2.64	1.69	1.69	0.09	0.002

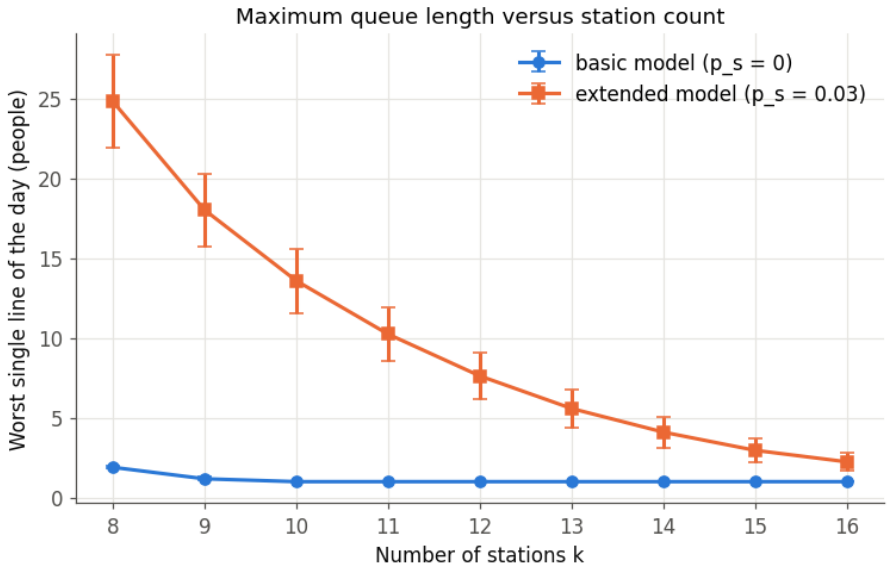


Figure 6: Mean of the daily worst single line against k for the basic and extended models, over 50 days, with 95% confidence intervals of the mean. This is the longest single line of the day, not the airport-wide total.

Routing explains the basic model completely. With random routing, the basic simulation matches the independent-line formula at every k (0.464 against 0.465 min at $k = 8$, 0.126 against 0.127 at $k = 16$). The formula is correct; it simply describes a different checkpoint. Shortest-line routing, at 0.035 min, falls between random routing and the pooled line (0.016 min), which is where it belongs: travelers choose well on arrival but cannot switch lines later.

With the officer, routing is still the largest factor. Random routing waits 3.3 times longer than shortest-line routing at $k = 10$ and 18 times longer at $k = 16$ (Figure 7). A traveler choosing the shortest line almost never joins a blocked station; one choosing at random often does. The remaining gap between theory and random routing comes from the officer model. For $k \geq 14$, case (a) matches random routing closely, but near the threshold both theoretical cases blow up too early. At $k = 8$ the time-average number of blocked stations is 1.91 ± 0.20 rather than the predicted 2.58, which means each referred traveler waits about 3.8 min for the officer, not 6.0. A blocked station cannot send another referral until it is released, so the officer's queue stays shorter than an M/G/1 queue with Poisson arrivals would suggest. The same correction explains why the backlog at $k = 6$ grows more slowly than predicted (Appendix B). Case (b) is too high throughout.

7 Recommendation

The model cannot choose a service target; that is a policy decision. As a working example we require the mean wait to be at most one minute, and we pick the smallest k whose 95% confidence interval of the mean lies entirely below that target. At $k = 10$ the mean alone is 1.17 min. At $k = 11$ it is 0.78 ± 0.20 min, with an upper end of 0.98. **We therefore recommend 11 stations.** Compared with the threshold minimum of 8, this cuts the mean wait from 3.0 to 0.8 min and the worst line of the day from about 25 to about 10 people. The interval concerns the mean over many

stations. Section 6 then uses a controlled experiment to explain the gap between simulation and theory.

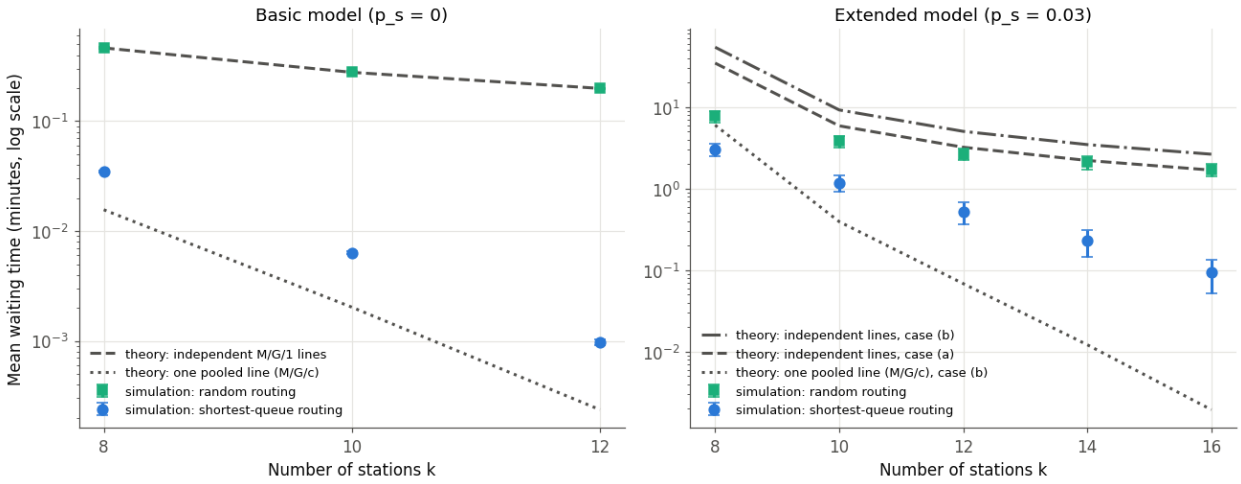


Figure 7: Mean waiting time against k on a log scale, for theory (gray lines) and for two simulations that differ only in how travelers choose a line (green squares: random line, 30 days; blue circles: shortest line, 50 days; bars are 95% confidence intervals of the mean). Left: basic model, shown up to $k = 12$, beyond which waits fall below 0.001 min. Right: extended model. Random routing reproduces the independent-line theory in the basic model, which isolates routing as the main cause of the gap.

days, not any single day, and the choice holds within the tested range of 8 to 16 stations rather than being a proven global optimum.

The choice is sensitive to what the airport cares about. If the worst line of the day should stay at or below five people, the same rule requires 15 stations (2.96 ± 0.77), since 14 stations give an upper end of 5.06. Beyond 11 stations the returns shrink quickly. At 14,400 travelers a day, the twelfth station saves about 62 traveler-hours per day, and the sixteenth only about 13. Staffing by the theory would be expensive: case (b) calls for 30 stations. The formulas locate the capacity threshold and the bottleneck well, but they should not choose the station count.

Finally, the officer is the real bottleneck. They are busy about 77% of the time regardless of k, and extra stations only soften the effect of their delays. A second senior officer would relieve the bottleneck directly, and it is the first thing we would test next.[9]

8 Conclusion and Limitations

A basic model of the checkpoint needs only 6 stations and produces almost no waiting. Adding one senior officer, who sees just 3% of travelers, raises the stable minimum to about 8 and accounts for almost all of the waiting that remains. For a one-minute mean wait we recommend 11 stations. Our main methodological lesson: the textbook formulas overestimated waits by up to a factor of 28, and calling them "approximate" explained nothing. Changing one assumption in the simulation showed that they are right for random line choice and that the gap comes from travelers choosing the shortest line.

The model is generous to the airport. Arrivals are constant over 24 hours, whereas real traffic peaks before morning flights and nearly stops at night, so our results describe a sustained peak of 10 travelers per minute. Travelers see line lengths perfectly and never switch lines; real behavior lies somewhere between our shortest-line and random-line cases, and Section 6 shows how much

[9] #composition: we state the recommendation and its justification first, then the conditions under which it changes, and keep caveats to the sentence where each applies.

that matters. Screening times are independent of the traveler, although a family with strollers is slow at every step. Finally, near the capacity threshold the queues fluctuate slowly and have heavy tails, so the evidence for stationarity at $k = 8$ and $k = 9$ is empirical, not a proof, and the stability of $k = 7$ remains open.

Acknowledgments & AI statement

The simulation engine builds on the `Event`, `Schedule`, and M/D/1 queue code from the CS166 pre-class workbooks and in-class sessions, and on the course's queueing-theory materials. We used Claude (Anthropic) to support in understanding and revision of the class and code concepts before starting this assignment. It was also used to help in reusing of the code from class and to help in debugging the simulation notebook. Additionally, it was also used to help in debugging Overleaf notebook and making sure the report is well-formatted. We reviewed and reran all code, checked the results against theory and the test cases, and made the modeling decisions and interpretations ourselves. Grammarly was used for proofreading.

References

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A Full Theoretical Derivations

This appendix gives every step of the analysis summarized in Section 4, in the same order: preliminary results, the four analytical steps, system-level capacity, and the limits of the analytical approach.

A.1 Notation, Units, and Preliminary Results

Travelers arrive according to a Poisson process with rate $\lambda = 10$ travelers per minute. Each traveler is flagged for additional screening with probability $p_s = 0.03$, and k denotes the number of service stations. Moments of the base screening time are computed in seconds, and moments of the additional screening time are computed in minutes. Every queueing formula is evaluated in minutes, so all service-time moments are converted to minutes before they are combined with λ.

Moments of a truncated normal distribution. Let $Y \sim \mathcal{N}(\mu, \sigma^2)$ and let X be Y conditioned on $Y > 0$, written $X \sim TN(\mu, \sigma^2; X > 0)$. Define the standardized truncation point

$$\alpha = \frac{0 - \mu}{\sigma} \tag{4}$$

and the inverse Mills ratio

$$\kappa = \frac{\phi(\alpha)}{1 - \Phi(\alpha)}, \tag{5}$$

where ϕ and Φ are the standard normal density and distribution function. We write κ rather than the more common λ to avoid a clash with the arrival rate. For the standardized variable $Z = (X - \mu)/\sigma$, integration by parts gives the recursion [2]

$$E[Z^j] = (j-1)\, E[Z^{j-2}] + \alpha^{j-1}\kappa, \tag{6}$$

from which

$$E[Z] = \kappa, \tag{7}$$
$$E[Z^2] = 1 + \alpha\kappa, \tag{8}$$
$$E[Z^3] = (\alpha^2 + 2)\,\kappa. \tag{9}$$

Since $X = \mu + \sigma Z$, the moments of X follow by binomial expansion:

$$E[X] = \mu + \sigma\kappa, \tag{10}$$
$$\mathrm{Var}(X) = \sigma^2 \left[1 + \alpha\kappa - \kappa^2\right], \tag{11}$$
$$E[X^2] = \mathrm{Var}(X) + E[X]^2, \tag{12}$$
$$E[X^3] = \mu^3 + 3\mu^2\sigma E[Z] + 3\mu\sigma^2 E[Z^2] + \sigma^3 E[Z^3]. \tag{13}$$

M/G/1 results. Consider an M/G/1 queue with Poisson arrival rate λ_i and generic service time S, with utilization

$$\rho_i = \lambda_i E[S] < 1. \tag{14}$$

The Pollaczek–Khinchine (P-K) formula gives the mean waiting time before service [4]:

$$W_q = \frac{\lambda_i E[S^2]}{2(1 - \rho_i)}. \tag{15}$$

Writing $\mathrm{CV}_S^2 = \mathrm{Var}(S)/E[S]^2$, the same formula can be expressed as

$$W_q = \frac{\rho_i}{1 - \rho_i} \cdot \frac{1 + \mathrm{CV}_S^2}{2} \cdot E[S], \tag{16}$$

which makes explicit that waiting time is driven jointly by utilization and service-time variability. Little's Law gives the mean number of travelers waiting:

$$L_q = \lambda_i W_q. \tag{17}$$

Approximation A1 (independent M/G/1 stations). We approximate each station as an independent M/G/1 queue fed by a Poisson stream of rate

$$\begin{aligned}\lambda_i &= \frac{\lambda}{k} \\ &= \frac{10}{k} \text{ travelers per minute.}\end{aligned} \tag{18}$$

This representation is exact if arrivals are assigned to stations uniformly at random, because independent random splitting of a Poisson process yields independent Poisson processes. Under the shortest-queue routing of the actual system, arrivals to a given station are neither Poisson nor independent of its state. Shortest-queue routing balances queue lengths and is therefore expected to reduce waiting times relative to random routing. We treat the resulting formulas as benchmarks, not as exact predictions.

Correspondence with simulated metrics. The assignment defines waiting time as the time between joining a queue and the start of screening, which corresponds exactly to W_q of the station queue. The total number of waiting travelers across all stations is

$$\begin{aligned}L_q^{\text{tot}} &= kL_q \\ &= \lambda W_q.\end{aligned} \tag{19}$$

The reported simulation metric is the average queue length per station, $L_q = \lambda W_q/k$, excluding travelers currently in service.

A.2 Basic Model: Regular Screening Only

The base screening time is $B \sim TN(30, 10^2; B > 0)$, measured in seconds. From (4):

$$\begin{aligned}\alpha_B &= \frac{0 - 30}{10} \\ &= -3.\end{aligned} \tag{20}$$

From (5):

$$\begin{aligned}\kappa_B &= \frac{\phi(-3)}{1 - \Phi(-3)} \\ &= \frac{0.0044318}{0.9986501} \\ &= 0.0044378.\end{aligned} \tag{21}$$

From (10):

$$\begin{aligned}E[B] &= 30 + 10(0.0044378) \\ &= 30.044378 \text{ s.}\end{aligned} \tag{22}$$

From (11):

$$\begin{aligned}\text{Var}(B) &= 100\left[1 + (-3)(0.0044378) - (0.0044378)^2\right] \\ &= 100\left[1 - 0.0133135 - 0.0000197\right] \\ &= 98.666679 \text{ s}^2.\end{aligned} \tag{23}$$

From (12):

$$\begin{aligned} E[B^2] &= 98.666679 + (30.044378)^2 \\ &= 98.666679 + 902.664673 \\ &= 1001.331352 \text{ s}^2. \end{aligned} \tag{24}$$

The squared coefficient of variation is

$$\begin{aligned} \text{CV}_B^2 &= \frac{98.666679}{902.664673} \\ &= 0.109306. \end{aligned} \tag{25}$$

Converting the mean to minutes:

$$\begin{aligned} E[B] &= \frac{30.044378}{60} \\ &= 0.500740 \text{ min}. \end{aligned} \tag{26}$$

Converting the second moment to minutes:

$$\begin{aligned} E[B^2] &= \frac{1001.331352}{3600} \\ &= 0.278148 \text{ min}^2. \end{aligned} \tag{27}$$

Capacity condition. Under A1, the utilization of each station is

$$\begin{aligned} \rho_i &= \lambda_i E[B] \\ &= \frac{10(0.500740)}{k} \\ &= \frac{5.007396}{k}. \end{aligned} \tag{28}$$

The requirement $\rho_i < 1$ is equivalent to

$$\begin{aligned} k &> \lambda E[B] \\ &= 5.007396, \end{aligned} \tag{29}$$

so the basic model requires

$$k_{\min} = 6. \tag{30}$$

Waiting time and queue length. Substituting (18), (27), and (28) into (15):

$$\begin{aligned} W_q^{(B)}(k) &= \frac{(10/k)(0.278148)}{2\,(1 - 5.007396/k)} \\ &= \frac{10(0.278148)}{2(k - 5.007396)} \\ &= \frac{1.390738}{k - 5.007396} \text{ min}. \end{aligned} \tag{31}$$

By (19):

$$L_q^{\mathrm{tot},(B)}(k) = 10\, W_q^{(B)}(k) = \frac{13.907380}{k - 5.007396}. \tag{32}$$

For example, at $k = 6$ the utilization is

$$\rho_i = \frac{5.007396}{6} = 0.834566, \tag{33}$$

the mean waiting time is

$$W_q^{(B)}(6) = \frac{1.390738}{0.992604} = 1.401 \text{ min}, \tag{34}$$

and the mean total number waiting is

$$L_q^{\mathrm{tot},(B)}(6) = 10(1.401) = 14.01 \text{ travelers}. \tag{35}$$

Because $\mathrm{CV}_B^2 \approx 0.11$, regular screening is nearly deterministic, so the basic model produces short waits even at a utilization of 83%.

A.3 Extended Model I: Additional Screening Without Officer Contention

This step answers the following question: how much service demand does additional screening add by itself, if the senior officer were always immediately available? The additional screening time is $A \sim TN(2, 2^2; A > 0)$, measured in minutes. From (4):

$$\alpha_A = \frac{0 - 2}{2} = -1. \tag{36}$$

From (5):

$$\kappa_A = \frac{\phi(-1)}{1 - \Phi(-1)} = \frac{0.2419707}{0.8413447} = 0.2876000. \tag{37}$$

From (10):

$$E[A] = 2 + 2(0.287600) = 2.575200 \text{ min} = 154.511997 \text{ s}. \tag{38}$$

From (11):

$$\begin{aligned}\mathrm{Var}(A) &= 4\left[1 + (-1)(0.287600) - (0.287600)^2\right] \\ &= 4\left[1 - 0.287600 - 0.082714\right] \\ &= 2.518745 \text{ min}^2 \\ &= 9067.482515 \text{ s}^2. \end{aligned} \tag{39}$$

From (12):

$$\begin{aligned}E[A^2] &= 2.518745 + (2.575200)^2 \\ &= 2.518745 + 6.631655 \\ &= 9.150400 \text{ min}^2 \\ &= 32941.439582 \text{ s}^2. \end{aligned} \tag{40}$$

The squared coefficient of variation is

$$\begin{aligned}\mathrm{CV}_A^2 &= \frac{2.518745}{6.631655} \\ &= 0.379806. \end{aligned} \tag{41}$$

Mixture service time. Let $I \sim \text{Bernoulli}(p_s)$ indicate whether a traveler is flagged, independent of B and A. The station service time when the officer is always free is

$$S_0 = B + IA. \tag{42}$$

Setting $p_s = 0$, or $\mu_2 = \sigma_2 = 0$, gives $S_0 = B$ and recovers the basic model. This is the analytical counterpart of the reduction check we ran on the simulation. Using linearity of expectation and $E[I] = p_s$:

$$\begin{aligned}E[S_0] &= E[B] + p_s E[A] \\ &= 30.044378 + 0.03(154.511997) \\ &= 30.044378 + 4.635360 \\ &= 34.679738 \text{ s}. \end{aligned} \tag{43}$$

Since $I^2 = I$ and I is independent of (B, A):

$$\begin{aligned}E[S_0^2] &= E[B^2] + 2p_s E[B]E[A] + p_s E[A^2] \\ &= 1001.331352 + 2(0.03)(30.044378)(154.511997) \\ &\quad + 0.03(32941.439582) \\ &= 1001.331352 + 278.533013 + 988.243187 \\ &= 2268.107553 \text{ s}^2. \end{aligned} \tag{44}$$

Consistency check with the two-component mixture. For a flagged traveler, assuming B and A are independent, the mean total time is

$$\begin{aligned}E[B + A] &= 30.044378 + 154.511997 \\ &= 184.556375 \text{ s}, \end{aligned} \tag{45}$$

the variance is

$$\begin{aligned}\mathrm{Var}(B+A) &= 98.666679 + 9067.482515 \\ &= 9166.149194 \text{ s}^2,\end{aligned} \tag{46}$$

and the second moment is

$$\begin{aligned}E[(B+A)^2] &= 9166.149194 + (184.556375)^2 \\ &= 9166.149194 + 34061.055517 \\ &= 43227.204711 \text{ s}^2.\end{aligned} \tag{47}$$

Mixing the two traveler types with weights 0.97 and 0.03:

$$\begin{aligned}E[S_0^2] &= 0.97(1001.331352) + 0.03(43227.204711) \\ &= 971.291411 + 1296.816141 \\ &= 2268.107552 \text{ s}^2,\end{aligned} \tag{48}$$

which agrees with (44) up to rounding.

Variance and variability.

$$\begin{aligned}\mathrm{Var}(S_0) &= 2268.107553 - (34.679738)^2 \\ &= 2268.107553 - 1202.684248 \\ &= 1065.423305 \text{ s}^2.\end{aligned} \tag{49}$$

$$\begin{aligned}\mathrm{CV}_{S_0}^2 &= \frac{1065.423305}{1202.684248} \\ &= 0.885871.\end{aligned} \tag{50}$$

Converting to minutes:

$$\begin{aligned}E[S_0] &= \frac{34.679738}{60} \\ &= 0.577996 \text{ min},\end{aligned} \tag{51}$$

$$\begin{aligned}E[S_0^2] &= \frac{2268.107553}{3600} \\ &= 0.630030 \text{ min}^2.\end{aligned} \tag{52}$$

Capacity and waiting time. The utilization of each station is

$$\begin{aligned}\rho_i &= \frac{10(0.577996)}{k} \\ &= \frac{5.779956}{k}.\end{aligned} \tag{53}$$

The requirement $\rho_i < 1$ is equivalent to

$$k > 5.779956, \tag{54}$$

so the minimum number of stations remains

$$k_{\min} = 6. \tag{55}$$

From (15):

$$\begin{aligned} W_q^{(S_0)}(k) &= \frac{10(0.630030)}{2(k - 5.779956)} \\ &= \frac{3.150149}{k - 5.779956} \text{ min.} \end{aligned} \tag{56}$$

At $k = 6$ the utilization is

$$\begin{aligned} \rho_i &= \frac{5.779956}{6} \\ &= 0.963326, \end{aligned} \tag{57}$$

and the mean waiting time is

$$\begin{aligned} W_q^{(S_0)}(6) &= \frac{3.150149}{0.220044} \\ &= 14.316 \text{ min.} \end{aligned} \tag{58}$$

Although only 3% of travelers are flagged, additional screening raises the mean occupation time by 15.4% and raises CV^2 from 0.109 to 0.886. At $k = 6$, the combined effect increases the predicted waiting time by a factor of about ten, from 1.401 to 14.316 minutes, while leaving the minimum station count unchanged.

A.4 Extended Model II: The Senior Officer as a Second Queueing System

The senior officer is a single shared server whose customers are the flagged travelers from all k stations. Before examining how the officer affects the stations, we analyze the officer as a queueing system in its own right.

Approximation A2 (Poisson arrivals to the officer). Independent thinning of a Poisson process with probability p_s yields a Poisson process with rate $p_s\lambda$. In the actual system, flagging occurs at the completion of base screening, so the officer's arrival stream is a thinned version of the station departure processes rather than of the external arrival process. In steady state its long-run rate is still $p_s\lambda$, and we approximate it as Poisson with rate

$$\begin{aligned} \lambda_s &= p_s\lambda \\ &= 0.03(10) \\ &= 0.3 \text{ flagged travelers per minute.} \end{aligned} \tag{59}$$

Utilization and stability. The officer's utilization is

$$\begin{aligned} \rho_s &= \lambda_s E[A] \\ &= 0.3(2.575200) \\ &= 0.772560, \end{aligned} \tag{60}$$

so that

$$1 - \rho_s = 0.227440. \tag{61}$$

The officer is stable because $\rho_s < 1$, and this condition does not depend on k. Although only 3% of travelers require the officer, each additional screening is on average

$$\begin{aligned} \frac{E[A]}{E[B]} &= \frac{2.575200}{0.500740} \\ &= 5.14 \end{aligned} \tag{62}$$

times as long as a base screening, so a single officer is loaded to 77.3% of capacity.

Mean waiting time for the officer (M/G/1). Because A follows a truncated normal distribution rather than an exponential one, we model the officer as an M/G/1 queue. Applying (15):

$$\begin{aligned} E[W_s] &= \frac{\lambda_s E[A^2]}{2(1-\rho_s)} \\ &= \frac{0.3(9.150400)}{2(0.227440)} \\ &= \frac{2.745120}{0.454880} \\ &= 6.034822 \text{ min.} \end{aligned} \tag{63}$$

Sensitivity to the service-time distribution (M/M/1). Replacing A by an exponential distribution with the same mean sets $\mathrm{CV}_A^2 = 1$ and gives the M/M/1 delay

$$\begin{aligned} E[W_s^{\mathrm{M/M/1}}] &= \frac{\rho_s E[A]}{1-\rho_s} \\ &= \frac{0.772560(2.575200)}{0.227440} \\ &= \frac{1.989497}{0.227440} \\ &= 8.747346 \text{ min.} \end{aligned} \tag{64}$$

By (16), the two delays differ only through the factor $(1 + \mathrm{CV}_A^2)/2$:

$$\begin{aligned} \frac{E[W_s]}{E[W_s^{\mathrm{M/M/1}}]} &= \frac{1 + 0.379806}{2} \\ &= 0.689903. \end{aligned} \tag{65}$$

As a check,

$$0.689903(8.747346) = 6.034822 \text{ min}, \tag{66}$$

which agrees with (63). Hence, with identical mean workload, the exponential assumption overstates the officer delay by 45%. We adopt the M/G/1 value (63) as the primary approximation and retain (64) as a sensitivity case.

Second moment of the officer delay. Treating W_s as a constant ignores its variability. For an M/G/1 queue under first-come, first-served discipline, the second moment of the waiting time is given by Takács' formula [5], which requires the third moment of the service time. From (7) with $\kappa_A = 0.287600$:

$$E[Z_A] = 0.287600. \tag{67}$$

From (8) with $\alpha_A = -1$:

$$\begin{aligned} E[Z_A^2] &= 1 - 0.287600 \\ &= 0.712400. \end{aligned} \tag{68}$$

From (9):

$$\begin{aligned} E[Z_A^3] &= (1+2)(0.287600) \\ &= 0.862800. \end{aligned} \tag{69}$$

From (13) with $\mu_2 = 2$ and $\sigma_2 = 2$:

$$\begin{aligned} E[A^3] &= 2^3 + 3(2^2)(2)(0.287600) \\ &\quad + 3(2)(2^2)(0.712400) + 2^3(0.862800) \\ &= 8 + 6.902399 + 17.097601 + 6.902399 \\ &= 38.902399 \text{ min}^3. \end{aligned} \tag{70}$$

Takács' formula gives

$$E[W_s^2] = 2E[W_s]^2 + \frac{\lambda_s E[A^3]}{3(1-\rho_s)}, \tag{71}$$

where the first term can equivalently be written as

$$2E[W_s]^2 = \frac{\lambda_s^2 (E[A^2])^2}{2(1-\rho_s)^2}. \tag{72}$$

Substituting the numerical values:

$$\begin{aligned} E[W_s^2] &= 2(6.034822)^2 + \frac{0.3(38.902399)}{3(0.227440)} \\ &= 72.838149 + \frac{11.670720}{0.682320} \\ &= 72.838149 + 17.104465 \\ &= 89.942615 \text{ min}^2. \end{aligned} \tag{73}$$

The variance of the officer delay is

$$\begin{aligned} \mathrm{Var}(W_s) &= 89.942615 - (6.034822)^2 \\ &= 89.942615 - 36.419075 \\ &= 53.523540 \text{ min}^2, \end{aligned} \tag{74}$$

so its standard deviation is

$$\begin{aligned} \sqrt{\mathrm{Var}(W_s)} &= \sqrt{53.523540} \\ &= 7.316 \text{ min}, \end{aligned} \tag{75}$$

which is larger than its mean. This result holds under the idealized M/G/1 assumption and is not an exact result for the coupled system.

Stations occupied by the officer subsystem. Applying Little's Law to the officer's waiting line gives the mean number of stations blocked while their flagged traveler waits for the officer:

$$\begin{aligned}\lambda_s E[W_s] &= 0.3(6.034822) \\ &= 1.810447.\end{aligned} \tag{76}$$

Applying Little's Law to the officer's service gives the mean number of stations held during additional screening:

$$\begin{aligned}\lambda_s E[A] &= \rho_s \\ &= 0.772560.\end{aligned} \tag{77}$$

These quantities reappear in the capacity decomposition below.

A.5 Effective Station Occupation Time with Blocking

Coupling assumption. A flagged traveler keeps the station occupied while waiting for the officer and while undergoing additional screening, so no other traveler in that queue can begin screening. The time for which a station is occupied by one traveler is therefore the random variable

$$S_{\text{eff}} = B + I\,(A + W_s), \tag{78}$$

where W_s is the flagged traveler's waiting time for the officer. This is the point at which the station queues and the officer queue become coupled.

Mean.

$$\begin{aligned}E[S_{\text{eff}}] &= E[B] + p_s\,(E[A] + E[W_s]) \\ &= 0.500740 + 0.03(2.575200 + 6.034822) \\ &= 0.500740 + 0.03(8.610022) \\ &= 0.500740 + 0.258301 \\ &= 0.759040 \text{ min.}\end{aligned} \tag{79}$$

Second moment. Using $I^2 = I$ and the independence of I from (B, A, W_s), and without yet assuming any other independence,

$$E[S_{\text{eff}}^2] = E[B^2] + 2p_s\,E[B(A + W_s)] + p_s\,E\big[(A + W_s)^2\big]\,. \tag{80}$$

Approximation A3 (independence). We assume that B is independent of (A, W_s). In an M/G/1 queue under first-come, first-served discipline, a customer's waiting time depends only on the customers ahead of it, so A and W_s are independent within A2. The cross term then factors as

$$\begin{aligned}E[B(A + W_s)] &= E[B]\,(E[A] + E[W_s]) \\ &= 0.500740(8.610022) \\ &= 4.311379 \text{ min}^2,\end{aligned} \tag{81}$$

and the squared term expands as

$$E\big[(A + W_s)^2\big] = E[A^2] + 2E[A]E[W_s] + E[W_s^2]. \tag{82}$$

The middle term of (82) is

$$\begin{aligned} 2E[A]E[W_s] &= 2(2.575200)(6.034822) \\ &= 31.081746 \text{ min}^2. \end{aligned} \tag{83}$$

The first two terms of (80) are common to both cases below:

$$\begin{aligned} 2p_s\, E[B(A+W_s)] &= 2(0.03)(4.311379) \\ &= 0.258683 \text{ min}^2. \end{aligned} \tag{84}$$

We evaluate $E[W_s^2]$ in (82) in two ways.

Case (a): deterministic delay. Replacing W_s by its mean, so that $E[W_s^2]$ is replaced by

$$\begin{aligned} E[W_s]^2 &= (6.034822)^2 \\ &= 36.419075 \text{ min}^2, \end{aligned} \tag{85}$$

the squared term is

$$\begin{aligned} E\big[(A+W_s)^2\big] &= 9.150400 + 31.081746 + 36.419075 \\ &= 76.651220 \text{ min}^2. \end{aligned} \tag{86}$$

The second moment of the effective service time is

$$\begin{aligned} E[S_{\text{eff}}^2] &= 0.278148 + 0.258683 + 0.03(76.651220) \\ &= 0.278148 + 0.258683 + 2.299537 \\ &= 2.836367 \text{ min}^2. \end{aligned} \tag{87}$$

The squared mean is

$$\begin{aligned} E[S_{\text{eff}}]^2 &= (0.759040)^2 \\ &= 0.576142 \text{ min}^2, \end{aligned} \tag{88}$$

so the variance is

$$\begin{aligned} \text{Var}(S_{\text{eff}}) &= 2.836367 - 0.576142 \\ &= 2.260225 \text{ min}^2, \end{aligned} \tag{89}$$

and the squared coefficient of variation is

$$\begin{aligned} \text{CV}_{\text{eff}}^2 &= \frac{2.260225}{0.576142} \\ &= 3.923. \end{aligned} \tag{90}$$

Case (b): random delay. Substituting the Takács second moment from (73), the squared term becomes

$$\begin{aligned} E\big[(A+W_s)^2\big] &= 9.150400 + 31.081746 + 89.942615 \\ &= 130.174760 \text{ min}^2. \end{aligned} \tag{91}$$

The second moment of the effective service time is

$$\begin{aligned} E[S_{\text{eff}}^2] &= 0.278148 + 0.258683 + 0.03(130.174760) \\ &= 0.278148 + 0.258683 + 3.905243 \\ &= 4.442073 \text{ min}^2. \end{aligned} \tag{92}$$

The variance is

$$\begin{aligned} \text{Var}(S_{\text{eff}}) &= 4.442073 - 0.576142 \\ &= 3.865931 \text{ min}^2, \end{aligned} \tag{93}$$

and the squared coefficient of variation is

$$\begin{aligned} \text{CV}_{\text{eff}}^2 &= \frac{3.865931}{0.576142} \\ &= 6.710. \end{aligned} \tag{94}$$

Randomness in the officer delay leaves $E[S_{\text{eff}}]$ unchanged but raises CV_{eff}^2 from 3.92 to 6.71. The shared resource therefore affects station performance through variability as well as through mean workload.

Sensitivity case: M/M/1 officer with deterministic delay. Substituting $c = E[W_s^{\text{M/M/1}}] = 8.747346$ into the case (a) calculation gives the M/M/1 sensitivity case. The mean is

$$\begin{aligned} E[S_{\text{eff}}] &= 0.500740 + 0.03(2.575200 + 8.747346) \\ &= 0.500740 + 0.03(11.322546) \\ &= 0.500740 + 0.339676 \\ &= 0.840416 \text{ min}. \end{aligned} \tag{95}$$

The cross term is

$$\begin{aligned} 2p_s\, E[B]\, (E[A] + c) &= 2(0.03)(0.500740)(11.322546) \\ &= 0.340179 \text{ min}^2. \end{aligned} \tag{96}$$

The squared term is

$$\begin{aligned} E\big[(A + c)^2\big] &= 9.150400 + 2(2.575200)(8.747346) + (8.747346)^2 \\ &= 9.150400 + 45.052327 + 76.516054 \\ &= 130.718781 \text{ min}^2. \end{aligned} \tag{97}$$

The second moment is

$$\begin{aligned} E[S_{\text{eff}}^2] &= 0.278148 + 0.340179 + 0.03(130.718781) \\ &= 0.278148 + 0.340179 + 3.921563 \\ &= 4.539890 \text{ min}^2. \end{aligned} \tag{98}$$

The variance is

$$\begin{aligned} \text{Var}(S_{\text{eff}}) &= 4.539890 - (0.840416)^2 \\ &= 4.539890 - 0.706299 \\ &= 3.833591 \text{ min}^2, \end{aligned} \tag{99}$$

and the squared coefficient of variation is

$$\begin{aligned} \mathrm{CV}_{\mathrm{eff}}^2 &= \frac{3.833591}{0.706299} \\ &= 5.428. \end{aligned} \tag{100}$$

A.6 System-Level Capacity and Staffing Analysis

Capacity decomposition. Under A1 with service time S_{eff}, the utilization of each station is

$$\rho_i = \frac{\lambda E[S_{\mathrm{eff}}]}{k}, \tag{101}$$

and the approximate capacity condition $\rho_i < 1$ becomes

$$\begin{aligned} k &> \lambda E[S_{\mathrm{eff}}] \\ &= \lambda E[B] + \lambda_s E[A] + \lambda_s E[W_s]. \end{aligned} \tag{102}$$

By Little's Law, the three terms are the mean numbers of stations engaged in base screening, held during additional screening, and blocked while waiting for the officer. Under the M/G/1 officer model:

$$\begin{aligned} \lambda E[S_{\mathrm{eff}}] &= 5.007396 + 0.772560 + 1.810447 \\ &= 7.590403, \end{aligned} \tag{103}$$

so that

$$k_{\min} = 8. \tag{104}$$

Under the M/M/1 sensitivity case, the officer-waiting term becomes

$$\begin{aligned} \lambda_s E[W_s^{\mathrm{M/M/1}}] &= 0.3(8.747346) \\ &= 2.624204, \end{aligned} \tag{105}$$

so that

$$\begin{aligned} \lambda E[S_{\mathrm{eff}}] &= 5.007396 + 0.772560 + 2.624204 \\ &= 8.404160, \end{aligned} \tag{106}$$

and

$$k_{\min} = 9. \tag{107}$$

Congestion at the officer therefore costs about 1.8 stations of capacity under the M/G/1 approximation, more than the 0.77 stations consumed by the additional screening itself.

Two coupled capacity constraints. The system is subject to two distinct capacity constraints. The officer requires

$$\lambda p_s E[A] < 1, \tag{108}$$

and the stations require

$$\lambda E[S_{\mathrm{eff}}] < k. \tag{109}$$

Constraint (108) does not involve k, so no number of regular stations can compensate for an overloaded officer. The constraints are also coupled: by (63), $E[W_s] \to \infty$ as $\rho_s \to 1$, so by (102) the station requirement grows without bound as the officer approaches saturation. At the given parameters $\rho_s = 0.773$, so the officer is stable but already contributes substantially to the station requirement.

Waiting time and queue length. Substituting $\lambda_i = \lambda/k$ and (101) into (15):

$$\begin{aligned} W_q(k) &= \frac{(\lambda/k)\, E[S_{\text{eff}}^2]}{2\,(1 - \lambda E[S_{\text{eff}}]/k)} \\ &= \frac{\lambda E[S_{\text{eff}}^2]}{2\,(k - \lambda E[S_{\text{eff}}])}. \end{aligned} \tag{110}$$

The associated queue-length and system-time measures are

$$L_q^{\text{tot}}(k) = \lambda W_q(k), \tag{111}$$

$$L_q(k) = \frac{\lambda W_q(k)}{k}, \tag{112}$$

$$W(k) = W_q(k) + E[S_{\text{eff}}]. \tag{113}$$

For case (a):

$$\begin{aligned} W_q(k) &= \frac{10(2.836367)}{2(k - 7.590403)} \\ &= \frac{14.181835}{k - 7.590403} \text{ min}. \end{aligned} \tag{114}$$

For case (b):

$$\begin{aligned} W_q(k) &= \frac{10(4.442073)}{2(k - 7.590403)} \\ &= \frac{22.210366}{k - 7.590403} \text{ min}. \end{aligned} \tag{115}$$

For the M/M/1 sensitivity case:

$$\begin{aligned} W_q(k) &= \frac{10(4.539890)}{2(k - 8.404160)} \\ &= \frac{22.699449}{k - 8.404160} \text{ min}. \end{aligned} \tag{116}$$

Worked example. Under case (a) at $k = 9$, the utilization is

$$\begin{aligned} \rho_i &= \frac{7.590403}{9} \\ &= 0.843378, \end{aligned} \tag{117}$$

the mean waiting time is

$$\begin{aligned} W_q(9) &= \frac{14.181835}{9 - 7.590403} \\ &= \frac{14.181835}{1.409597} \\ &= 10.061 \text{ min}, \end{aligned} \tag{118}$$

the mean total number waiting is

$$\begin{aligned} L_q^{\text{tot}}(9) &= 10(10.061) \\ &= 100.61 \text{ travelers}, \end{aligned} \tag{119}$$

and the mean time in system is

$$\begin{aligned} W(9) &= 10.061 + 0.759 \\ &= 10.820 \text{ min.} \end{aligned} \tag{120}$$

Table 3 in Section 4 and Tables 6 and 7 below summarize the progression across the analytical steps.

Table 6: Approximate mean waiting time $W_q(k)$ in minutes. "Unstable" means that the approximate capacity condition fails.

k	Regular	Officer free	M/G/1 (a)	M/G/1 (b)	M/M/1
6	1.401	14.316	unstable	unstable	unstable
7	0.698	2.582	unstable	unstable	unstable
8	0.465	1.419	34.624	54.225	unstable
9	0.348	0.978	10.061	15.757	38.097
10	0.279	0.746	5.886	9.217	14.224
11	0.232	0.603	4.159	6.514	8.745
12	0.199	0.506	3.216	5.037	6.313
13	0.174	0.436	2.622	4.106	4.939
14	0.155	0.383	2.213	3.465	4.056
15	0.139	0.342	1.914	2.998	3.441
16	0.127	0.308	1.686	2.641	2.988

Table 7: Approximate total number of waiting travelers $L_q^{\text{tot}}(k) = \lambda W_q(k)$ across all queues. "Unstable" means that the approximate capacity condition fails.

k	Regular	Officer free	M/G/1 (a)	M/G/1 (b)	M/M/1
6	14.01	143.16	unstable	unstable	unstable
7	6.98	25.82	unstable	unstable	unstable
8	4.65	14.19	346.24	542.25	unstable
9	3.48	9.78	100.61	157.57	380.97
10	2.79	7.46	58.86	92.17	142.24
11	2.32	6.03	41.59	65.14	87.45
12	1.99	5.06	32.16	50.37	63.13

Analytical station counts. Applying the one-minute target of Section 7 to these formulas gives the smallest k with $W_q(k) \leq 1$ minute: 7 for regular screening only, 9 with the officer always free, 22 for case (a), 30 for case (b), and 32 for the M/M/1 sensitivity case.

A.7 Maximum Queue Length and Limits of the Analytical Approach

The analysis above addresses three kinds of questions with different degrees of rigor. Mean waiting time is obtained from the P-K formula under approximations A1 to A3. Mean queue length then follows from Little's Law, which holds for any stable system and adds no further approximation. The maximum queue length over a day is an extreme-value statistic of a dependent stochastic process, and we do not derive a closed form for it. Instead, we identify the mechanisms that govern it.

First, for a single M/G/1 queue in heavy traffic, the waiting time is approximately exponentially distributed with mean given by (15) (Kingman's heavy-traffic approximation [3]). Upper quantiles of the queue length therefore scale approximately with

$$\frac{\rho_i \left(1 + \mathrm{CV}_S^2\right)}{1 - \rho_i}, \tag{121}$$

so the maximum should fall steeply as k increases beyond the capacity threshold and should be much larger than the mean when $\mathrm{CV}_{\text{eff}}^2$ lies between 3.9 and 6.7. Second, when k is below the approximate threshold (102), the total backlog grows over the day (Appendix A.8 derives the rate), so both the daily average and the daily maximum depend on the length of the observation horizon rather than on a stationary distribution. Third, officer blocking can temporarily freeze several stations at once, which produces bursts of queue growth that are correlated across stations and that no single-queue formula captures.

The main limitations of the analytical model are as follows.

1. **Routing (A1).** The per-station formulas are exact under random routing. Shortest-queue routing balances the queues and reduces waiting times, so the analytical values overestimate the simulated waits. Section 6 confirms this directly: a simulation with random routing reproduces the independent-queue formula for the basic model.

2. **Officer arrivals (A2).** Flagged travelers reach the officer through the station departure processes, not as an independent Poisson stream. Moreover, at most k travelers can wait for the officer, and a blocked station cannot produce a new flagged traveler until it is released. This finite-source feedback limits the officer's queue and makes (63) an overestimate, particularly for small k. In the simulation at $k = 8$, the implied officer wait is about 3.8 min rather than 6.0 (Section 6).

3. **Independence (A3).** The factorizations in (81) and (82) assume that the base screening time is independent of the officer delay, and the per-station model treats different stations as independent even though they share the officer.

4. **Steady state.** The formulas describe stationary behavior in the long run. Each simulated day, by contrast, covers a finite horizon of 1440 minutes after a warm-up from an empty system.

In summary, within its stated approximations the theory shows that regular screening alone requires at least 6 stations. Additional screening, if the officer were always available, raises service variability substantially but does not change this requirement. Contention for the single senior officer raises the approximate requirement to 8 stations under the M/G/1 officer model and to 9 under the M/M/1 sensitivity case, with officer waiting accounting for most of this increase. The officer's own capacity constraint cannot be relaxed by adding regular stations.

A.8 Growth rate below the threshold

If $k < \lambda E[S_{\text{eff}}]$, the total backlog grows. The naive rate $\lambda - k/E[S_{\text{eff}}]$ assumes that the officer still receives λp_s referrals per minute, which cannot happen when the stations are overloaded. If only a throughput $\theta < \lambda$ gets through, the officer receives $p_s\theta$ referrals per minute and waits less. We therefore solve

$$\theta\, h(\theta) = k, \qquad h(\theta) = E[B] + p_s \left(E[A] + W_s(p_s\theta)\right), \tag{122}$$

where $W_s(\cdot)$ is the M/G/1 officer wait at that referral rate. The backlog then grows at $\lambda - \theta$ (Table 9).

A.9 Pooled line

For a single line feeding c stations, with offered load $r = \lambda E[S]$, we use the Allen–Cunneen approximation [1]

$$W_q^{M/G/c} \approx C(c,r)\,\frac{E[S]}{c-r}\cdot\frac{1+\mathrm{CV}_S^2}{2}, \qquad C(c,r) = \frac{\dfrac{r^c}{c!\,(1-r/c)}}{\displaystyle\sum_{j=0}^{c-1}\frac{r^j}{j!} + \frac{r^c}{c!\,(1-r/c)}}, \tag{123}$$

where $C(c,r)$ is the Erlang C probability of waiting.

B Detailed Empirical Evidence

B.1 Single-station check

With one station and no officer, the simulated mean queue length matches the Pollaczek–Khinchine formula at every tested utilization (30 runs of 3,000 min each, after a 200-minute warm-up): 0.071 against 0.071 at $\rho = 0.3$, 0.274 against 0.277 at $\rho = 0.5$, 0.883 against 0.906 at $\rho = 0.7$, 1.732 against 1.775 at $\rho = 0.8$, and 4.450 against 4.493 at $\rho = 0.9$, with the theory inside every 95% confidence interval (notebook §5.4).

B.2 Stationarity of the test cases

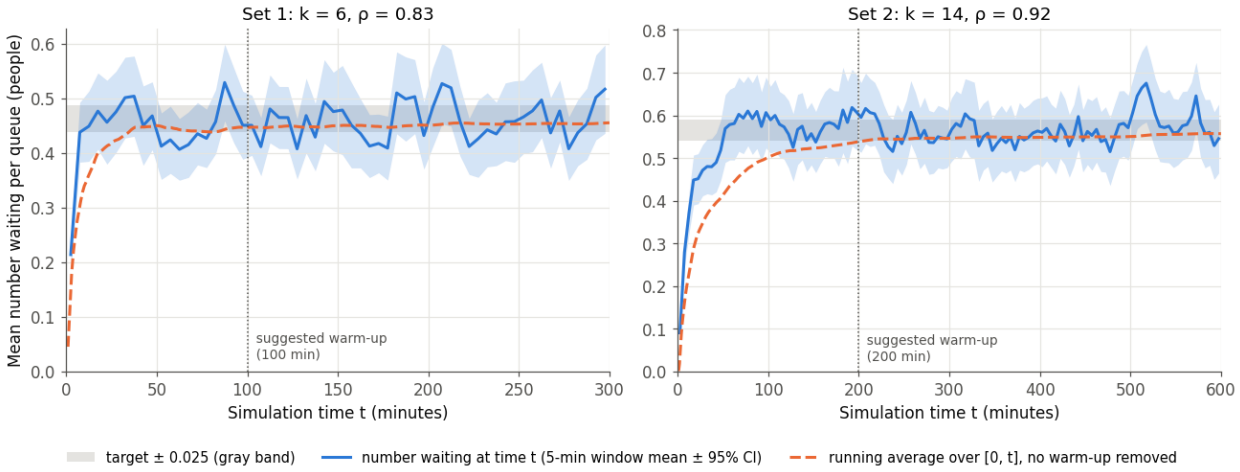


Figure 8: Transient behavior of the two test cases, averaged over 200 runs from an empty airport. Blue: ensemble mean number waiting per line in 5-minute windows, with 95% confidence band. Orange dashed: running average from $t = 0$, which keeps the start-up bias. Gray band: target ± 0.025. Dotted line: suggested warm-up.

For set 1, all 50-minute windows lie between 0.443 and 0.467 and their intervals overlap. For set 2, the first 100-minute window (0.500) lies clearly below the later ones (0.555–0.592), which overlap one another. The running average from $t = 0$ is still below the plateau at 200 minutes for set 2, which is why the warm-up must be discarded rather than averaged in.

B.3 Warm-up near the capacity threshold

Table 8: Ensemble mean number waiting per line in consecutive blocks, extended model, runs from an empty airport, with 95% confidence intervals of the mean. The last column is an independent replication with new seeds.

Block (min)	$k = 8$ (100 runs)	$k = 9$ (100 runs)	$k = 9$ (200 new runs)
$[0, 100)$	1.62 ± 0.55	0.67 ± 0.25	0.65 ± 0.18
$[100, 200)$	3.44 ± 0.99	1.76 ± 0.81	1.57 ± 0.41
$[200, 300)$	4.30 ± 1.43	3.45 ± 1.16	1.89 ± 0.50
$[300, 500)$	4.08 ± 1.29	2.03 ± 0.77	2.23 ± 0.55
$[500, 700)$	3.02 ± 0.88	2.14 ± 0.61	2.28 ± 0.49
$[700, 900)$	4.28 ± 1.18	1.53 ± 0.40	1.90 ± 0.51

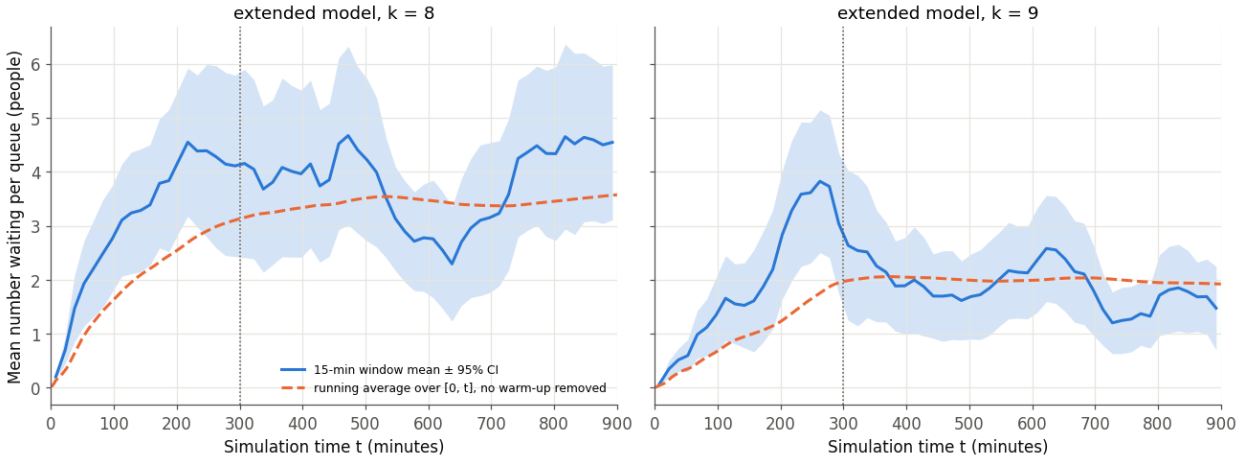


Figure 9: Transient of the extended model at $k = 8$ and $k = 9$, averaged over 100 runs from an empty airport. Blue: ensemble mean in 15-minute windows with 95% confidence band; orange dashed: running average from $t = 0$; dotted line: the 300-minute warm-up.

In both cases the first 100 minutes lie well below the later blocks. At $k = 9$ the first set of runs shows a bump at 200–300 minutes, but it does not reproduce: with 200 new seeds, the mean rises steadily to about 2 and levels off by 300 minutes. The bump came from heavy tails. The five largest of the original 100 runs carry 35% of that block's total, and the median run is far below the mean in every block (0.43 against 1.89 at $[200, 300)$ in the replication). After 300 minutes the blocks overlap and agree with the day-long mean of 1.98 ± 0.39 measured in the sweep, so we use a 300-minute warm-up. Heavy tails are also why we average 50 independent days rather than relying on one long run.

B.4 Stability below the threshold

The throughput-consistent prediction halves the gap at $k = 6$; the rest comes from the M/G/1 officer approximation discussed in Section 6. At $k = 7$ a growth of 0.41 per minute is not visible within 1,000 minutes, so we cannot yet say whether seven stations are stable. The question does not affect our recommendation.

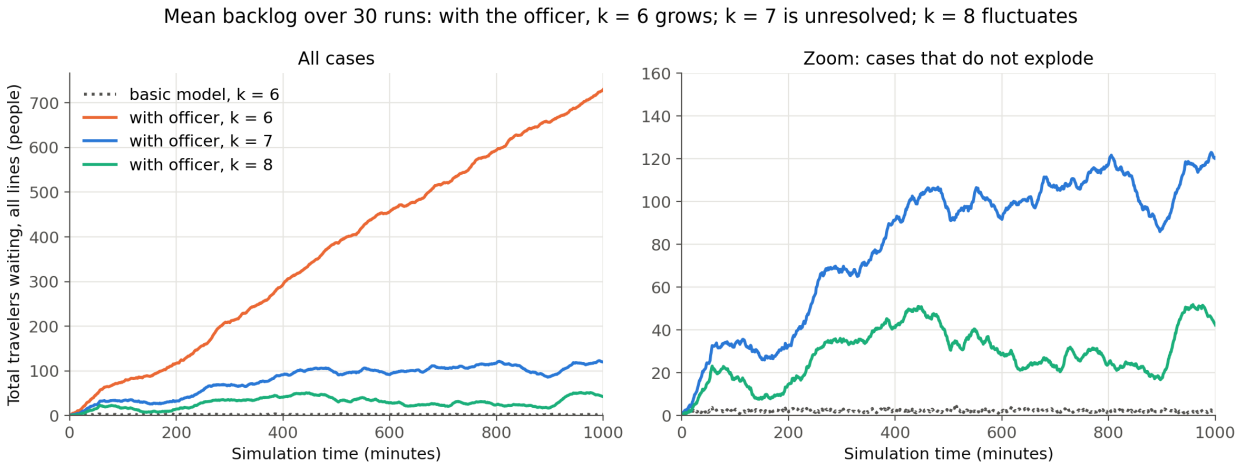


Figure 10: Mean total number waiting across all lines over 1,000 minutes, averaged over 30 runs from an empty airport. Right panel: zoom on the cases that do not grow rapidly. Without the officer, 6 stations settle almost immediately; with the officer, 6 stations grow steadily, 7 drift upward slowly, and 8 fluctuate without a trend.

Table 9: Growth rate of the total backlog (travelers per minute): naive and throughput-consistent predictions against the slope of the simulated ensemble mean between 300 and 1,000 minutes (30 runs per k).

k	Naive $\lambda - k/E[S_{\text{eff}}]$	Throughput-consistent	Simulated
6	$+2.10$	$+1.28$	$+0.72$
7	$+0.78$	$+0.41$	0.00
8	-0.54	-0.54	$+0.07$

B.5 Mechanisms behind the worst line

The worst line falls from 6.6 to 38 times the mean queue between $k = 8$ and $k = 16$. The heavy-traffic scale $\rho(1 + \mathrm{CV}^2)/(1 - \rho)$ from Kingman's result [3], evaluated with the case (b) moments, falls from 143 to 7 over the same range: it captures the direction and steepness of the decline, not the headcount. At $k = 8$, the eight lines move together (mean pairwise correlation 0.96 ± 0.03). This is consistent with one officer freezing several stations at once, and equally with shortest-queue routing, which moves arrivals between lines; the correlation alone does not separate the two. Either way, a single-queue formula has no term for a shock shared across lines.

B.6 Confidence intervals

Each replication contributes one value x_r of each metric. The sample mean is

$$\bar{x} = \frac{1}{R} \sum_{r=1}^{R} x_r, \tag{124}$$

the sample variance is

$$s^2 = \frac{1}{R-1} \sum_{r=1}^{R} (x_r - \bar{x})^2, \tag{125}$$

and the 95% confidence interval of the mean is

$$\bar{x} \pm t_{0.975,\, R-1} \frac{s}{\sqrt{R}}, \tag{126}$$

with $t_{0.975,\, 49} = 2.010$ for $R = 50$. Because the replications use distinct seeds, the x_r are independent and identically distributed. The interval describes uncertainty about the mean over days. It is not a prediction interval for a single day.

B.7 Little's law consistency check

Because waiting time and queue length are measured independently in the code, Little's law in the form (112) provides a consistency check between the two metrics. At $k = 8$ it predicts

$$\begin{aligned} L_q(8) &= \frac{10}{8} (2.988) \\ &= 3.735, \end{aligned} \tag{127}$$

against a measured 3.736. At $k = 12$ it predicts

$$\begin{aligned} L_q(12) &= \frac{10}{12} (0.522) \\ &= 0.435, \end{aligned} \tag{128}$$

against a measured 0.436. The agreement confirms that the two metrics are recorded consistently. Since Little's law holds for any stable system, it does not by itself validate the queueing dynamics.

Table 10: Mean waiting time (minutes): independent-line theory against the shortest-line simulation for every k in the sweep. The last column is the ratio of case (b) to the extended simulation.

	Regular screening		With officer			
k	Theory	Simulation	Case (a)	Case (b)	Simulation	Ratio
8	0.465	0.035	34.624	54.225	2.988	18.1
9	0.348	0.015	10.061	15.757	1.782	8.8
10	0.279	0.006	5.886	9.217	1.174	7.9
11	0.232	0.003	4.159	6.514	0.781	8.3
12	0.199	0.001	3.216	5.037	0.522	9.6
13	0.174	<0.001	2.622	4.106	0.346	11.9
14	0.155	<0.001	2.213	3.465	0.228	15.2
15	0.139	<0.001	1.914	2.998	0.147	20.4
16	0.127	<0.001	1.686	2.641	0.093	28.4

B.8 Full comparison of theory and simulation

B.9 Theory and simulation: detailed observations

Table 10 supports four observations.

1. **The qualitative structure agrees.** Theory and simulation agree that the officer, not regular screening, drives congestion; that the officer's own utilization is about 0.77 regardless of k, as the simulated 0.778 ± 0.007 confirms; and that waiting times fall steeply above the approximate threshold near 7.6 stations.

2. **The analytical values overestimate the simulated waits at every tested k.** Case (b) exceeds the extended simulation by a factor between 7.9 and 28.4. Case (a), which ignores the variance of the officer delay, is lower but still 5 to 18 times the simulated value. Because the M/G/1 benchmarks are not bounds for the coupled system, these gaps are empirical findings rather than a confirmation of a theoretical inequality.

3. **The gap is not a constant factor.** The absolute gap narrows as k grows, from 51.2 minutes at $k = 8$ to 2.5 minutes at $k = 16$, but the ratio is smallest near $k = 10$ and rises on both sides. Near the boundary the $1/(k - 7.59)$ singularity of (115) dominates. For large k the simulated wait approaches zero while the benchmark decays only as $1/k$.

4. **Routing accounts for most of the gap.** Without the officer, the independent-queue formula exceeds the simulation by a factor of 13 at $k = 8$ (0.465 against 0.035 minutes). The random-routing experiment of Section 6 closes this gap completely, so in the basic model shortest-queue routing is the whole explanation. With the officer, routing explains most of the gap and the officer approximations the rest.